

Homework 1

Isabel Moreau Linear Algebra Sep 18, 2026 GRADE: 29/40 (Prof. R. Castellano)

Problem 1

(a)

$$2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$2u = [4, -2, 6]^T$$

$$2u - v = [4 - 1, -2 - 4, 6 - (-2)]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

$$[\text{ANSWER: } (2u - v) \cdot w = 24]$$

[PROF: ✓]

(b)

$$(1/2)w = [-2, 1, 3]^T$$

$$v + (1/2)w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$||v + (1/2)w|| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3}$$

$$[\text{ANSWER: } 3\sqrt{3} \approx 5.20]$$

[PROF: ✓]

(c)

$$\cos \theta = (u \cdot w) / (||u|| ||w||)$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$||u|| = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$||w|| = \sqrt{16 + 4 + 36} = \sqrt{56}$$

$$||w|| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$||u|| ||w|| = \sqrt{14} \cdot 2\sqrt{14} = 28$$

$$\cos \theta = 8/28 = 2/7$$

$$\theta = \arccos(2/7)$$

[ANSWER: $\theta \approx 73.40^\circ$]

[PROF: ✓]

[PROF: Problem 1 5/5]

Problem 2

(a)

AB is undefined, because A is 3×3 (3 columns) and B is 2×3 (2 rows), and $3 \neq 2$.

BA is 2×3 since B is 2×3 and A is 3×3 .

~~$$\text{row 1: } [1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(1),$$~~

$$\text{row 1: } [1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$$

$$\text{row 2: } [3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$$

$$[\text{ANSWER: } BA = [2 \ -9 \ -2]$$

$$[7 \ 4 \ -1]]$$

[PROF: ✓]

(b)

AV is $(3 \times 3)(3 \times 1) = 3 \times 1$.

$$AV = [2(1) + (-1)(-2) + 0(3)] \quad [4]$$

$$[1(1) + 3(-2) + (-2)(3)] = [-11]$$

$$[0(1) + 4(-2) + 1(3)] \quad [5]$$

VA is undefined: v is 3×1 so it has 1 column, A has 3 rows, and $1 \neq 3$.

$$[\text{ANSWER: } AV = [4, -11, 5]^T, VA \text{ undefined}]$$

[PROF: -0.5 E8 Third entry of AV is $0(1) + 4(-2) + 1(3) = -5$, not 5.]

(c)

Aw is undefined: A has 3 columns and w is 1×3 with only 1 row, $3 \neq 1$.

$$wA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)]$$

$$[ANSWER: wA = [4 \ -6 \ 1], Aw \text{ undefined}]$$

[PROF: -0.5 E8 Third entry of wA is $2(0) + 0(-2) + (-1)(1) = -1$, not 1.]

(d)

vw is $(3 \times 1)(1 \times 3) = 3 \times 3$, so each entry is a v entry times a w entry.

$$vw = [1(2) \ 1(0) \ 1(-1)] [2 \ 0 \ -1]$$

$$[-2(2) \ -2(0) \ -2(-1)] = [-4 \ 0 \ 2]$$

$$[3(2) \ 3(0) \ 3(-1)] [6 \ 0 \ -3]$$

wv is $(1 \times 3)(3 \times 1) = 1 \times 1$ and equals the dot product: $2(1) + 0(-2) + (-1)(3) = -1$

$$[ANSWER: vw \text{ as above, } wv = [-1]]$$

[PROF: ✓]

(e)

$A^2 = AA$, defined since A is square.

$$\text{row 1: } [2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$$

$$\text{row 2: } [1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$$

$$\text{row 3: } [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 12, -7]$$

$$[ANSWER: A^2 = [3 \ -5 \ 2]$$

$$[5 \ 0 \ -8]$$

$$[4 \ 12 \ -7]]]$$

[PROF: -0.5 E8 Entry (3,2) of A^2 is $0(-1) + 4(3) + 1(4) = 16$, not 12.]

(f)

B^2 undefined.

[PROF: -0.5 E5 Undefined with no dimension reason; state that B is 2×3 , so BB would need $3 = 2$.]

[PROF: Problem 2 4/6]

Problem 3

(a)

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1 \quad \begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & 0 \end{bmatrix}$$

$$R_2 \rightarrow -R_2 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2 \quad \begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 12 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 6/5 \end{bmatrix}$$

$$x_3 = 6/5$$

$$x_2 = -4 + 3(6/5) = -4 + 18/5 = -2/5$$

$$x_1 = -1 - 2(-2/5) + 6/5 = -1 + 4/5 + 6/5 = 1$$

$$[\text{ANSWER: } x = [1, -2/5, 6/5]^T]$$

[PROF: -1 E8 In $R_3 \rightarrow R_3 + R_1$ the constant is $-1 + (-1) = -2$, not 0; the elimination and back substitution after the slip follow your numbers correctly.]

(b)

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$[0 \ 1 \ 3 \mid -2]$$

$$R_3 \rightarrow R_3 - R_2 \quad [1 \ 0 \ -7 \mid 8]$$

$$R_1 \rightarrow R_1 - 2R_2 \quad [0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

No pivot in column 3, so z is free. Let $z = t$. Then $y = -2 - 3t$ and $x = 8 + 7t$, so the system has infinitely many solutions.

$$\text{ANSWER: } [x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}$$

[PROF: ✓]

[PROF: Problem 3 5/6]

Problem 4

(a)

Let $x_1 =$ kg of Alloy 1 used, $x_2 =$ kg of Alloy 2 used, $x_3 =$ kg of Alloy 3 used.

The 100 kg product has 36 kg copper, 44 kg zinc, and 20 kg nickel, and no metal is

lost, so for each metal the mass in equals the mass in the product.

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

[PROF: ✓]

(b)

Multiply everything by 10:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

[PROF: ✓]

(c)

$$R_1 \rightarrow R_1 - 2R_3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1 \quad [1 \ 0 \ -4 \mid -40]$$

$$R_3 \rightarrow R_3 - 2R_1 \quad [0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$~~R_2 \rightarrow (1/2)R_2~~$$

$$R_2 \rightarrow (1/7)R_2 \quad [1 \ 0 \ -4 \mid -40]$$

$$R_3 \rightarrow R_3 - R_2 \quad [0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$$R_3 \rightarrow (1/10)R_3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

$$x_3 = 20, x_2 = 80 - 2(20) = 40, x_1 = -40 + 4(20) = 40$$

Check: $40 + 40 + 20 = 100$ kg total, copper $20 + 8 + 8 = 36$ kg.

[ANSWER: 40 kg of Alloy 1, 40 kg of Alloy 2, 20 kg of Alloy 3]

[PROF: ✓]

[PROF: Problem 4 8/8]

Problem 5

(a)

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2 \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\text{[ANSWER: RREF}(A) = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}]$$

[PROF: ✓]

(b)

$\text{rank}(A) = 2$, since the RREF has two pivots (columns 1 and 3) and two nonzero rows.

$$\text{[ANSWER: rank}(A) = 2]$$

[PROF: ✓]

(c)

0 solutions: possible.

1 solution: impossible.

Infinitely many: possible.

[PROF: -4 E7 Three verdicts with no reason for any of them; a claim about the number of solutions earns no credit until you connect it to an inconsistent row, to the free variable in column 2, or to rank $2 < 3$ unknowns.]

[PROF: Problem 5 3/7]

Problem 6

(a)

These are dependent because there are three vectors and v_3 has a zero entry, so they can't fill $\mathbb{R}^3$.

[ANSWER: dependent, span is a plane]

[PROF: -2 E7 A zero entry in v_3 says nothing about dependence; the justification points need the row reduction with two pivots or the relation $v_3 = v_2 - 2v_1$.]

(b)

None of these is a multiple of another so they are independent.

[ANSWER: independent, span is $\mathbb{R}^3$]

[PROF: -2 E7 Pairwise non-multiplicity does not prove independence; show three pivots or $\det = 2 \neq 0$.]

[PROF: Problem 6 4/8]

Grader summary

Problem 1: 5/5

Problem 2: 4/6

Problem 3: 5/6

Problem 4: 8/8

Problem 5: 3/7

Problem 6: 4/8

Total: 29/40

Your elimination is clean, every row operation is labeled in the right form, and Problems 3(b) and 4 are fully correct. Three entry slips in Problem 2 and one constant slip in 3(a) — $R_3 \rightarrow R_3 + R_1$ gives -2, not 0 — cost you the arithmetic points. The real loss is justification: 5(c) states three verdicts with no reasons at all, and both parts of Problem 6 rest on arguments that prove nothing, so row reduce and count pivots or exhibit the dependence relation.